

Free Module

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Let R be a ring with $1 \neq 0$.

1 Internal Freeness

Definition 1.1. (*internal free module of a set*)

Let F be an R -module, and let $A \subseteq F$.

We say that F is **internal free on A** , if $\forall x \in F$, $\exists!$ function $c_x : A \rightarrow R$ with finite support s.t.

$$x = \sum_{a \in \text{supp}(c_x)} c_x(a) \cdot_F a.$$

In this case, A is called a **basis** (or a **set of free generators**) for F .

If R is commutative and F is free on some basis A , then the cardinality of A is called the **rank of F** .

Lemma 1.1. (*internal: well-behavior with $\emptyset$*)

Let R be a ring with $1 \neq 0$, let F be an R -module, and let $A \subseteq F$.

Suppose that F is internally free on A . Then

$$A = \emptyset \iff F = 0.$$

Proof. First suppose that $A = \emptyset$. Since F is free on A , every element of F is a finite R -linear combination of elements of A . But there are no elements of A , so the only such linear combination is the empty sum, which is 0. Hence every element of F is equal to 0, and therefore

$$F = 0.$$

Conversely, suppose that $F = 0$. We show that $A = \emptyset$. Assume for contradiction that $A \neq \emptyset$, and choose $a \in A$.

Since F is free on A , the element $0 \in F$ must admit a unique representation by a finitely supported function $c : A \rightarrow R$ such that

$$0 = \sum_{x \in \text{supp}(c)} c(x) x.$$

Now let $c_0 : A \rightarrow R$ be the zero function. Then c_0 has finite support, and

$$0 = \sum_{x \in \text{supp}(c_0)} c_0(x) x.$$

Define another function $c_1 : A \rightarrow R$ by

$$c_1(a) = 1, \quad c_1(x) = 0 \text{ for all } x \in A \setminus \{a\}.$$

Since $A \subseteq F$ and $F = 0$, we have $a = 0$. Hence

$$\sum_{x \in \text{supp}(c_1)} c_1(x) x = 1 \cdot a = 1 \cdot 0 = 0.$$

So c_1 is also a finitely supported function representing 0.

But $c_1 \neq c_0$, since

$$c_1(a) = 1 \neq 0 = c_0(a).$$

□

Theorem 1.1. (*Theorem 10.6*)

Let A be an arbitrary set, R be a ring with $1 \neq 0$.

Then, there is an R -module $F(A)$ internally free on $\iota(A)$, where $\iota : A \rightarrow F(A)$, s.t. $F(A)$ satisfies the **universal property**.

Universal property:

Let M be any R -module, $\varphi : A \rightarrow M$ be any map of sets.

Then, $\exists!$ R -module homomorphism $\Phi : F(A) \rightarrow M$ s.t.

$$\Phi \circ \iota = \varphi$$

If A is finite i.e. $A = \{a_1, \dots, a_n\}$ for $n \geq 1$, then $F(A) = Ra_1 \oplus \dots \oplus Ra_n \cong R^n$.

2 External Freeness

Definition 2.1. (*external free module of a set*)

Let A be an arbitrary set, R be a ring with $1 \neq 0$.

An **external free module on A** is a R -module $\mathcal{F}(A)$ equipped with an **inclusion function** $\iota : A \rightarrow \mathcal{F}(A)$ that satisfies the **universal property**.

Universal property:

Let M be any R -module, $\varphi : A \rightarrow M$ be any map of sets.

Then, $\exists!$ R -module homomorphism $\Phi : \mathcal{F}(A) \rightarrow M$ s.t.

$$\Phi \circ \iota = \varphi$$

Lemma 2.1. (*injectivity of the inclusion function*)

Let A be an arbitrary set, R be a ring with $1 \neq 0$.

Let $\mathcal{F}(A)$ be an external free R -module on A , $\iota : A \rightarrow \mathcal{F}(A)$ be the equipped inclusion map.

Then, ι is injective.

Proof intuition: map into canonical free module.

Corollary 2.0.1. (*Corollary 10.7 (1)*)

Let R be a ring with $1 \neq 0$.

If (F_1, ι_1) and (F_2, ι_2) are two external free R -modules on the same set A , then there exists a unique R -module isomorphism

$$\Psi : F_1 \rightarrow F_2$$

such that

$$\Psi \circ \iota_1 = \iota_2.$$

Lemma 2.2. (*external: well-behavior with $\emptyset$*)

Let R be a ring with $1 \neq 0$, A be a set. Let $(F, \iota : A \rightarrow F)$ be an external free module on A . Then,

$$A = \emptyset \iff F = 0.$$

Proof intuition: prove Corollary 10.7(1) first

3 Bridge

Lemma 3.1. (*isomorphism preserves internal freeness*)

Let R be a ring with $1 \neq 0$, let M_1, M_2 be R -modules, $A_1 \subseteq M_1$.

Suppose A_1 is internally free on M_1 .

Suppose $\Phi : M_1 \rightarrow M_2$ is an R -module isomorphism.

Then, $\Phi(A_1)$ is internally free on M_2 .

Theorem 3.1. (*bridge theorem 1 : external $\implies$ internal*)

Let R be a ring with $1 \neq 0$, A be a set. Let $(F, \iota : A \rightarrow F)$ be an external free module on A .

Then, F is an internal free module on $\iota(A)$.

Proof. First, assume $A \neq \emptyset$, otherwise trivial.

We have two external free modules on A : canonical $(\mathcal{F}, \iota_c)$ and given (F, ι) .

By Corollary 10.7 (1), we have a unique R -module isomorphism $\Phi : \mathcal{F} \rightarrow F$ s.t.

$$\Phi \circ \iota_c = \iota$$

We also know that $\iota_c(A)$ is internally free on $\mathcal{F}$. Apply previous lemma, we have

$$\Phi(\iota_c(A)) = \iota(A)$$

is internally free on F . □

Theorem 3.2. (*bridge theorem 2 : internal $\implies$ external*)

Let R be a ring with $1 \neq 0$, F be an internal free module on $A \subseteq F$.

Then, $(F, \text{inc} : A \rightarrow F)$ is an external free module on A .

Proof. Let M be any R -module, and let $\varphi : A \rightarrow M$ be any map of sets. We must show that there exists a unique R -module homomorphism $\Phi : F \rightarrow M$ such that

$$\Phi(\text{inc}(a)) = \varphi(a), \quad \forall a \in A.$$

Since F is internally free on A , for each $x \in F$ there exists a unique finitely supported function $c_x : A \rightarrow R$ such that

$$x = \sum_{a \in \text{supp}(c_x)} c_x(a) a.$$

Define a map $\Phi : F \rightarrow M$ by

$$\Phi(x) := \sum_{a \in \text{supp}(c_x)} c_x(a) \varphi(a).$$

This is well-defined because the coefficient function c_x is unique for each $x \in F$.

We claim that Φ is an R -module homomorphism. Let $x, y \in F$, and let $r \in R$. Write

$$x = \sum_{a \in \text{supp}(c_x)} c_x(a) a, \quad y = \sum_{a \in \text{supp}(c_y)} c_y(a) a.$$

Then

$$x + y = \sum_{a \in \text{supp}(c_x) \cup \text{supp}(c_y)} (c_x(a) + c_y(a)) a,$$

where we interpret $c_x(a) = 0$ for $a \notin \text{supp}(c_x)$, and similarly for c_y . By uniqueness of coefficients, the coefficient

function of $x + y$ is $c_x + c_y$. Hence

$$\Phi(x + y) = \sum_{a \in \text{supp}(c_x) \cup \text{supp}(c_y)} (c_x(a) + c_y(a)) \varphi(a) = \Phi(x) + \Phi(y).$$

Similarly,

$$rx = \sum_{a \in \text{supp}(c_x)} (r c_x(a)) a,$$

so by uniqueness of coefficients, the coefficient function of rx is $r c_x$. Therefore

$$\Phi(rx) = \sum_{a \in \text{supp}(c_x)} (r c_x(a)) \varphi(a) = r \sum_{a \in \text{supp}(c_x)} c_x(a) \varphi(a) = r \Phi(x).$$

Thus Φ is an R -module homomorphism.

Now for each $a \in A$, the element $a \in F$ has coefficient function $c_a : A \rightarrow R$ given by

$$c_a(a) = 1, \quad c_a(b) = 0 \text{ for all } b \in A \setminus \{a\}.$$

Hence, $\Phi(\text{inc}(a)) = \Phi(a) = 1 \cdot \varphi(a) = \varphi(a)$. So Φ extends φ .

It remains to prove uniqueness. Let $\Psi : F \rightarrow M$ be any R -module homomorphism such that

$$\Psi(\text{inc}(a)) = \varphi(a), \quad \forall a \in A.$$

Take any $x \in F$, and write

$$x = \sum_{a \in \text{supp}(c_x)} c_x(a) a.$$

Then

$$\Psi(x) = \Psi\left(\sum_{a \in \text{supp}(c_x)} c_x(a) a\right) = \sum_{a \in \text{supp}(c_x)} c_x(a) \Psi(a) = \sum_{a \in \text{supp}(c_x)} c_x(a) \varphi(a) = \Phi(x).$$

Therefore $\Psi = \Phi$.

We have shown that for every R -module M and every set map $\varphi : A \rightarrow M$, there exists a unique R -module homomorphism $\Phi : F \rightarrow M$ such that $\Phi \circ \text{inc} = \varphi$. Hence $(F, \text{inc} : A \rightarrow F)$ is an external free module on A . $\square$

Corollary 3.2.1. (Corollary 10.7 (2))

Let R be a ring with $1 \neq 0$.

Let F be an R -module, and let $A \subseteq F$. If F is internally free on A , and if $(F(A), \iota)$ is an external free R -module on the set A , then there exists an R -module isomorphism

$$\Phi : F(A) \rightarrow F$$

such that

$$\Phi(\iota(a)) = a, \quad \forall a \in A.$$

In particular, (F, inc_A) enjoys the same universal property with respect to A as $F(A)$ does.

4 General Freeness

Definition 4.1. (*general free modules, free generators, rank of a free module*)

Let R be a ring with $1 \neq 0$.

An R -module F is a **free module** if there exists a set A such that

$$F \cong \mathcal{F}(A),$$

where $(\mathcal{F}(A), \iota)$ is an external free R -module on A .

A is called an **external basis** for F .

If R is commutative, the cardinality of A is called the **external rank** of F .

Lemma 4.1. (*alternative formulation of a free module*)

Let R be a ring with $1 \neq 0$, M be an R -module. Then, **T.F.A.E.**

1. M is free
2. $\exists B \subseteq M$ s.t. M is internally free on B

4.1 Excursion: Linear Algebra

Theorem 4.1. (*Every spanning set contains a basis*)

Let F be a field, V be a vector space over F , and let $S \subseteq V$.
If S spans V , then S contains a basis of V .

Proof. Let $\mathcal{A} := \{L \subseteq S : L \text{ is linearly independent}\}$.
Then $\mathcal{A} \neq \emptyset$, since $\emptyset \in \mathcal{A}$.

We apply Zorn's lemma in the following form: if $\mathcal{X}$ is a nonempty set of sets such that for every chain $\mathcal{C} \subseteq \mathcal{X}$ under $\subseteq$, one has $\bigcup \mathcal{C} \in \mathcal{X}$, then $\mathcal{X}$ has a maximal element under $\subseteq$.

So let $\mathcal{B} \subseteq \mathcal{A}$ be a chain under $\subseteq$. We claim that

$$\bigcup \mathcal{B} \in \mathcal{A}.$$

Set $L^* := \bigcup \mathcal{B}$. We show that L^* is linearly independent.

Let $n \in \omega$, let $v : n \rightarrow L^*$ be injective, and let $\alpha : n \rightarrow F$ satisfy

$$\sum_{i \in n} \alpha(i) v(i) = 0_V.$$

Since n is finite, the set $v[n]$ is finite. For each $x \in v[n]$, choose $B_x \in \mathcal{B}$ such that $x \in B_x$. Because $\mathcal{B}$ is a chain and $v[n]$ is finite, there exists $B_0 \in \mathcal{B}$ such that

$$B_x \subseteq B_0 \quad \text{for all } x \in v[n].$$

Hence, $v[n] \subseteq B_0$.

Since $B_0 \in \mathcal{A}$, the set B_0 is linearly independent. Therefore

$$\forall i \in n, \alpha(i) = 0_F.$$

So L^* is linearly independent, and thus

$$\bigcup \mathcal{B} \in \mathcal{A}.$$

By Zorn's lemma, $\mathcal{A}$ has a maximal element; let $B \in \mathcal{A}$ be maximal. Then $B \subseteq S$ and B is linearly independent. We claim that B spans V .

Suppose not. Then

$$\text{Span}(B) \neq V.$$

Since S spans V , not every element of S can lie in $\text{Span}(B)$. Otherwise

$$S \subseteq \text{Span}(B),$$

and hence

$$V = \text{Span}(S) \subseteq \text{Span}(B),$$

a contradiction.

So there exists

$$s \in S \setminus \text{Span}(B).$$

We show that $B \cup \{s\}$ is linearly independent.

Let $n \in \omega$, let $v : n \rightarrow B \cup \{s\}$ be injective, and let $\alpha : n \rightarrow F$ satisfy

$$\sum_{i \in n} \alpha(i) v(i) = 0_V.$$

If $s \notin v[n]$, then $v[n] \subseteq B$, so by linear independence of B ,

$$\forall i \in n, \alpha(i) = 0_F.$$

Now suppose $s \in v[n]$. Since v is injective, there is a unique $j \in n$ such that

$$v(j) = s.$$

Then

$$\alpha(j)s + \sum_{i \in n, i \neq j} \alpha(i) v(i) = 0_V.$$

If $\alpha(j) \neq 0_F$, then

$$s = -\alpha(j)^{-1} \sum_{i \in n, i \neq j} \alpha(i) v(i) \in \text{Span}(B),$$

contrary to the choice of s . Therefore

$$\alpha(j) = 0_F.$$

Hence

$$\sum_{i \in n, i \neq j} \alpha(i) v(i) = 0_V,$$

and since all $v(i)$ for $i \neq j$ lie in B , linear independence of B gives

$$\forall i \in n, i \neq j, \alpha(i) = 0_F.$$

So again

$$\forall i \in n, \alpha(i) = 0_F.$$

Thus $B \cup \{s\}$ is linearly independent. Since $B \cup \{s\} \subseteq S$, we have

$$B \cup \{s\} \in \mathcal{A},$$

contradicting the maximality of B .

Therefore B spans V . Since B is both linearly independent and spanning, B is a basis of V . Also $B \subseteq S$. Hence S contains a basis of V . $\square$

Theorem 4.2. (Every L.I. set can be extended to a basis)

Let F be a field, V be a vector space over F , and let $S \subseteq V$ be a linearly independent set of vectors. Then, $\exists B \subseteq V$ s.t.

$$S \subseteq B \quad \text{and} \quad B \text{ is a basis of } V$$

Proof. Assume $V \neq 0$ and $S \neq \emptyset$.

Let

$$\mathcal{A}_0 := \{L \subseteq V : S \subseteq L \text{ and } L \text{ is linearly independent}\},$$

and define

$$\mathcal{A} := \mathcal{A}_0 \cup \{\emptyset\}.$$

We order $\mathcal{A}$ by $\subseteq$. We verify the hypothesis of Zorn's lemma in Enderton's form.

Let $\mathcal{B} \subseteq \mathcal{A}$ be a chain. We show that

$$\bigcup \mathcal{B} \in \mathcal{A}.$$

If $\mathcal{B} = \emptyset$, then

$$\bigcup \mathcal{B} = \emptyset \in \mathcal{A}.$$

Now suppose $\mathcal{B} \neq \emptyset$. If every element of $\mathcal{B}$ is $\emptyset$, then again

$$\bigcup \mathcal{B} = \emptyset \in \mathcal{A}.$$

So suppose $\mathcal{B}$ contains some nonempty L_0 . Then $L_0 \in \mathcal{A}_0$, hence

$$S \subseteq L_0 \subseteq \bigcup \mathcal{B}.$$

It remains to show that $\bigcup \mathcal{B}$ is linearly independent. Let

$$v_1, \dots, v_n \in \bigcup \mathcal{B}$$

be distinct, and suppose

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0.$$

For each i , choose $L_i \in \mathcal{B}$ such that $v_i \in L_i$. Since $\mathcal{B}$ is a chain and $\{L_1, \dots, L_n\}$ is finite, there exists $L \in \mathcal{B}$ such that

$$L_1, \dots, L_n \subseteq L.$$

Then

$$v_1, \dots, v_n \in L.$$

Also $L \neq \emptyset$, so $L \in \mathcal{A}_0$, and therefore L is linearly independent. Hence

$$\alpha_1 = \dots = \alpha_n = 0.$$

Thus $\bigcup \mathcal{B}$ is linearly independent, and therefore

$$\bigcup \mathcal{B} \in \mathcal{A}_0 \subseteq \mathcal{A}.$$

By Zorn's lemma, $\mathcal{A}$ has a maximal element B .

Since S is linearly independent and $S \neq \emptyset$, we have

$$S \in \mathcal{A}_0 \subseteq \mathcal{A}$$

and

$$\emptyset \subsetneq S.$$

Hence $B \neq \emptyset$. So $B \in \mathcal{A}_0$. In particular,

$$S \subseteq B \quad \text{and} \quad B \text{ is linearly independent.}$$

We claim that B spans V . Suppose not. Then choose

$$v \in V \setminus \text{Span}(B).$$

We show that $B \cup \{v\}$ is linearly independent.

Suppose

$$\alpha v + \beta_1 b_1 + \cdots + \beta_m b_m = 0$$

with $b_1, \dots, b_m \in B$. If $\alpha \neq 0$, then

$$v = -\alpha^{-1}(\beta_1 b_1 + \cdots + \beta_m b_m) \in \text{Span}(B),$$

a contradiction. Hence $\alpha = 0$. Then

$$\beta_1 b_1 + \cdots + \beta_m b_m = 0,$$

and since B is linearly independent, all $\beta_i = 0$. Therefore $B \cup \{v\}$ is linearly independent.

Since

$$S \subseteq B \subseteq B \cup \{v\},$$

we have

$$B \cup \{v\} \in \mathcal{A}_0 \subseteq \mathcal{A}.$$

But

$$B \subsetneq B \cup \{v\},$$

contradicting the maximality of B .

Therefore B spans V . Hence B is a basis of V , and $S \subseteq B$. □

Theorem 4.3. (invariant of dimension on vector spaces)

Let F be a field, let V be a vector space over F , let $S \subseteq V$ be a spanning set of V , and let $I \subseteq V$ be a linearly independent set. Then

$$|I| \leq |S|.$$

In particular, any two bases of V have the same cardinality.

Proof. By the theorem that every linearly independent subset of a vector space is contained in a basis, there exists a basis

$$B \subseteq V$$

such that

$$I \subseteq B.$$

For each $s \in S$, since B is a basis of V , there exists a unique finitely supported function

$$c_s : B \rightarrow F$$

such that

$$s = \sum_{b \in \text{supp}(c_s)} c_s(b) b.$$

Set

$$E_s := \text{supp}(c_s) \subseteq B.$$

Then each E_s is finite.

Now let

$$E := \bigcup_{s \in S} E_s.$$

Then

$$E \subseteq B.$$

We claim that E spans V . Indeed, let $v \in V$. Since S spans V , there exist pairwise distinct elements

$$s_1, \dots, s_n \in S$$

and scalars

$$\alpha_1, \dots, \alpha_n \in F$$

such that

$$v = \alpha_1 s_1 + \dots + \alpha_n s_n.$$

For each $k = 1, \dots, n$, we have

$$s_k = \sum_{b \in E_{s_k}} c_{s_k}(b) b.$$

Substituting these expressions into the displayed equation for v , we obtain an expression of v as a finite linear combination of elements of

$$E_{s_1} \cup \dots \cup E_{s_n} \subseteq E.$$

Hence

$$v \in \text{Span}(E).$$

Since $v \in V$ was arbitrary, E spans V .

But $E \subseteq B$, and B is a basis of V , hence a linearly independent spanning set. Therefore no proper subset of B can span V . Since E spans V , it follows that

$$E = B.$$

In particular,

$$I \subseteq B = E = \bigcup_{s \in S} E_s.$$

Now define

$$T := \{(s, b) \in S \times I : b \in E_s\}.$$

Since each $E_s \cap I$ is finite, for every $s \in S$ the fiber

$$T_s := \{b \in I : (s, b) \in T\} = E_s \cap I$$

is finite.

Moreover, since

$$I \subseteq \bigcup_{s \in S} E_s,$$

for every $b \in I$ there exists some $s \in S$ such that $b \in E_s$. Hence the projection

$$\pi_I : T \rightarrow I, \quad \pi_I(s, b) = b$$

is surjective.

By the axiom of choice, for each $s \in S$ choose a bijection

$$\varphi_s : T_s \rightarrow n_s$$

for some finite ordinal $n_s \in \omega$. Then the map

$$\Phi : T \rightarrow \bigcup_{s \in S} (\{s\} \times n_s), \quad \Phi(s, b) = (s, \varphi_s(b))$$

is a bijection. Therefore

$$|T| = \left| \bigcup_{s \in S} (\{s\} \times n_s) \right|.$$

Since each n_s is finite, we have

$$\left| \bigcup_{s \in S} (\{s\} \times n_s) \right| \leq |S \times \omega|.$$

If S is finite, then $|T|$ is finite and certainly $|T| \leq |S|$. If S is infinite, then

$$|S \times \omega| = |S|.$$

Hence in all cases

$$|T| \leq |S|.$$

Since there is a surjection

$$\pi_I : T \rightarrow I,$$

it follows that

$$|I| \leq |T| \leq |S|.$$

Thus

$$|I| \leq |S|.$$

Now let B_1 and B_2 be two bases of V . Since B_1 is linearly independent and B_2 spans V , the first part gives

$$|B_1| \leq |B_2|.$$

Similarly,

$$|B_2| \leq |B_1|.$$

Therefore, by Cantor–Bernstein,

$$|B_1| = |B_2|.$$

So any two bases of V have the same cardinality. □

4.2 Justify: rank of a free module

Theorem 4.4. (*invariance of basis cardinality for free modules over a commutative ring*)

Let R be a commutative ring with $1 \neq 0$, let M be an R -module, and let $A, B \subseteq M$.

Suppose that M is internally free on both A and B . Then

$$|A| = |B|.$$

Proof. Since $1 \neq 0$, the ring R is nonzero. Hence there exists a maximal ideal $\mathfrak{m} \subset R$. Set

$$k := R/\mathfrak{m}.$$

Then k is a field.

Now consider the quotient module

$$V := M/\mathfrak{m}M.$$

Since $\mathfrak{m}$ acts trivially on V , this is naturally a vector space over k .

Let

$$\pi : M \rightarrow V$$

be the quotient map. We will show that $\pi(A)$ and $\pi(B)$ are both bases of the k -vector space V .

First, we show that $\pi(A)$ spans V . Let $v \in V$. Choose $x \in M$ such that $\pi(x) = v$. Since M is internally free on A , there exists a unique finitely supported function

$$c_x : A \rightarrow R$$

such that

$$x = \sum_{a \in \text{supp}(c_x)} c_x(a) a.$$

Applying π , we obtain

$$v = \pi(x) = \sum_{a \in \text{supp}(c_x)} \overline{c_x(a)} \pi(a),$$

where $\overline{c_x(a)}$ denotes the class of $c_x(a)$ in $k = R/\mathfrak{m}$. Thus $\pi(A)$ spans V .

Next, we show that $\pi(A)$ is linearly independent over k . Let $a_1, \dots, a_n \in A$ be distinct, and suppose

$$\lambda_1 \pi(a_1) + \dots + \lambda_n \pi(a_n) = 0$$

for some $\lambda_1, \dots, \lambda_n \in k$. Choose $r_1, \dots, r_n \in R$ such that

$$\lambda_i = \overline{r_i} \quad (1 \leq i \leq n).$$

Then

$$\pi(r_1 a_1 + \dots + r_n a_n) = 0,$$

so

$$r_1 a_1 + \dots + r_n a_n \in \mathfrak{m}M.$$

Hence there exist $m_1, \dots, m_t \in \mathfrak{m}$ and $x_1, \dots, x_t \in M$ such that

$$r_1 a_1 + \dots + r_n a_n = \sum_{j=1}^t m_j x_j.$$

Since M is internally free on A , each x_j has a unique expression

$$x_j = \sum_{a \in \text{supp}(c_j)} c_j(a) a$$

for some finitely supported function $c_j : A \rightarrow R$. Substituting these expressions into the previous equation, we obtain

$$r_1 a_1 + \cdots + r_n a_n = \sum_{a \in F} s(a) a$$

for some finite set $F \subseteq A$, where every coefficient $s(a)$ lies in $\mathfrak{m}$.

Moving all terms to one side, we get a finite linear combination of elements of A equal to 0. Since A is a basis of M , all coefficients must vanish. In particular, for each i we obtain

$$r_i \in \mathfrak{m}.$$

Therefore

$$\lambda_i = \overline{r_i} = 0 \quad (1 \leq i \leq n).$$

Thus $\pi(A)$ is linearly independent over k .

So $\pi(A)$ is a basis of the vector space V . By the same argument, $\pi(B)$ is also a basis of V .

We now show that the restriction

$$\pi|_A : A \rightarrow \pi(A)$$

is injective. Suppose $a, a' \in A$ and

$$\pi(a) = \pi(a').$$

Then

$$a - a' \in \mathfrak{m}M.$$

So there exist $m_1, \dots, m_t \in \mathfrak{m}$ and $x_1, \dots, x_t \in M$ such that

$$a - a' = \sum_{j=1}^t m_j x_j.$$

Expanding each x_j uniquely in the basis A , the right-hand side becomes a finite linear combination of elements of A whose coefficients all lie in $\mathfrak{m}$. Hence we obtain an equality

$$a - a' = \sum_{u \in F} s(u) u$$

for some finite set $F \subseteq A$, where each $s(u) \in \mathfrak{m}$.

Move all terms to one side. Then we obtain a finite linear combination of elements of A equal to 0, in which the coefficient of a is $1 - s(a)$. Since $s(a) \in \mathfrak{m}$, we have $1 - s(a) \notin \mathfrak{m}$, for otherwise

$$1 = (1 - s(a)) + s(a) \in \mathfrak{m},$$

contrary to $\mathfrak{m} \neq R$. Thus $1 - s(a) \neq 0$, contradicting the linear independence of A unless $a = a'$. Therefore $\pi|_A$ is injective.

It is surjective onto $\pi(A)$ by definition, so $\pi|_A$ is a bijection. Hence

$$|A| = |\pi(A)|.$$

Similarly,

$$|B| = |\pi(B)|.$$

Since $\pi(A)$ and $\pi(B)$ are both bases of the vector space V , the invariance of dimension for vector spaces gives

$$|\pi(A)| = |\pi(B)|.$$

Therefore

$$|A| = |\pi(A)| = |\pi(B)| = |B|.$$

This completes the proof. □

5 Collected Facts

Theorem 5.1. (*same rank entails isomorphism*)

Let R be a ring with $1 \neq 0$, let M_1, M_2 be free R -modules of ranks r_1, r_2 (arbitrary cardinals).
If $r_1 = r_2$, then $M_1 \cong M_2$.

Note: converse to bridge lemma 1